

Autonomous Auction Bidding Agents

Nur Nuhaa Rusydah Jaffar
 School of Computer Science
 University of Nottingham
 efynj3@nottingham.ac.uk

Abstract—This project investigates the behaviour of autonomous bidding agents in a simulated multi-agent auction market. The environment is based on repeated first-price auction rounds, where agents compete to win items while attempting to maximise profit. Five bidding strategies were implemented and compared: Random, Conservative, Aggressive, Sniper and Adaptive. The Adaptive Agent uses auction memory to adjust its bidding aggressiveness based on previous outcomes, while the other agents follow fixed rule-based strategies. Four experiments were conducted to evaluate strategy performance, increasing competition, information availability and market noise. The results show that no single strategy dominated every metric. In the strategy comparison experiment, the Conservative Agent achieved the highest average profit, while the Adaptive Agent achieved the highest win rate but lower efficiency. Memory access also improved the Adaptive Agent's win rate but reduced its profit and efficiency by encouraging more competitive bidding. Increasing the number of agents sharply reduced average profit and bidding efficiency, while market noise created both opportunities and instability. Overall, the findings show that autonomous bidding agents should be evaluated using profit, win rate, efficiency and robustness together, rather than judging success by win rate alone.

Keywords—autonomous agents, multi-agent systems, auction simulation, bidding strategies, adaptive agents, agent memory, market noise, first-price auctions.

I. INTRODUCTION

Autonomous agents are increasingly used in environments that require independent decision-making under uncertain and competitive conditions. These systems can perceive information from their environment, process relevant data, and select actions that aim to maximise performance according to defined objectives [1]. One important application area for autonomous agents is auction-based decision-making, where agents compete against one another by placing bids to acquire resources or maximise profit [2], [4].

This project investigates the behaviour of autonomous bidding agents operating within a simulated auction market. The simulation is based on repeated auction rounds in which multiple agents compete to acquire generated items by submitting bids according to their individual strategies. Different autonomous bidding behaviours are implemented and evaluated, including rule-based strategies and an adaptive agent that uses previous auction outcomes to adjust its behaviour over time.

The aim of the project is to examine how different bidding strategies perform under changing market conditions. By comparing agent profitability, win rate, bidding efficiency and failure rates, the project evaluates how strategy design, competition, information availability and market noise affect the performance of autonomous agents.

II. BACKGROUND

Auction-based systems are widely used in digital and real-world markets, including online advertising, electronic

commerce, cloud resource allocation and spectrum licensing [4], [5]. In these environments, autonomous bidding agents must often make decisions under uncertainty, with limited information and dynamic competition from other agents [2], [6], [7]. The effectiveness of a bidding strategy can therefore significantly influence an agent's profitability, success rate and long-term performance within the market.

This project investigates the following questions:

1. How do different bidding strategies influence the profitability and performance of autonomous agents?
2. How does agent performance change under different market conditions, such as increased competition?
3. Can an adaptive agent improve its bidding behaviour over time by learning from previous auction outcomes?
4. To what extent does access to additional market information affect agent decision-making and performance?
5. How does market noise affect bidding performance and strategy stability?

The results of the simulation are analysed using repeated experimental runs, statistical summaries and visualisations. These are used to evaluate profitability, bidding efficiency, win rate, performance stability and adaptive behaviour across different auction conditions.

III. RELATED LITERATURE

Autonomous agents are commonly described as systems that perceive their environment and select actions in pursuit of defined goals [1]. In a multi-agent system, the outcome of one agent's decision is also influenced by the actions of other agents, making the environment more dynamic and competitive [2]. This is relevant to auction markets because each bidding agent must make decisions while considering not only the value of the item, but also the possible behaviour of other competing agents.

Auction theory provides the economic foundation for this project. Vickrey's work on auctions and competitive sealed tenders is an important early contribution to the formal study of auction mechanisms [3]. Auction theory has since been extended to study different auction formats, bidder incentives and strategic behaviour [4]. In a first-price auction, the winner pays the amount that they bid, meaning that bidders must balance the desire to win against the need to avoid overpaying. This trade-off directly relates to the performance measures used in this project, such as total profit, win rate and bidding efficiency.

The use of software agents in electronic commerce has also been widely studied. Guttman, Moukas and Maes describe how agent-mediated electronic commerce can automate activities such as product search, comparison, negotiation and purchasing decisions [5]. This supports the use of autonomous agents in simulated market environments,

where decision-making can be tested without human intervention. Similarly, the Trading Agent Competition demonstrated how autonomous bidding agents could be evaluated in competitive market scenarios involving bidding strategy, market prediction and resource allocation [6], [7]. These studies show that simulated markets are useful for analysing agent behaviour under uncertainty and competition.

Learning is another important concept in intelligent agent design. Reinforcement learning is based on the idea that agents can improve their behaviour through feedback from previous actions and outcomes [8]. Although this project does not implement a full reinforcement learning algorithm, the Adaptive Agent uses a simpler memory-based mechanism to adjust its aggressiveness based on previous auction results. This allows the project to explore whether limited learning behaviour can improve bidding performance compared with fixed rule-based strategies.

Overall, the literature suggests that autonomous bidding agents are suitable for studying decision-making in competitive environments. Previous work on auction theory, multi-agent systems and agent-mediated electronic commerce provides the foundation for this project. However, this project focuses on a smaller controlled simulation, allowing different bidding strategies to be compared under specific conditions such as competition level, information availability and market noise.

IV. METHODOLOGY

The auction market was tested using a set of autonomous bidding agents, each following a different decision-making strategy. These agents were designed to represent different levels of risk, competitiveness, timing behaviour, and access to market information, which are common considerations in auction-based multi-agent environments [2], [4], [6]. By comparing these strategies under the same auction environment, the experiments can evaluate how different agent behaviours affect profit, win rate, and bidding efficiency.

A. Random Agent

The Random Agent was used as a baseline strategy. It does not follow a fixed bidding pattern or use previous auction information to improve its decisions. Instead, it generates a random valid bid within an allowed range based on the current highest bid, its available balance, and the perceived value of the item.

This agent is useful because it provides a simple comparison point for the other strategies. Since its decisions are unpredictable, any strategy that performs consistently better than the Random Agent can be considered more effective than a basic non-strategic approach. However, the Random Agent may sometimes perform well by chance, especially in auctions where competition is low or where item values allow a wider range of profitable bids.

B. Conservative Agent

The Conservative Agent follows a low-risk bidding strategy. It only increases the current highest bid by a small amount and avoids making large bids close to the perceived item value. This allows the agent to reduce the chance of overpaying and preserve its balance over repeated auction rounds.

The main advantage of this strategy is that it can achieve good profit when it wins because it tends to pay less for items. However, its cautious behaviour also means that it may lose more auctions when competing against agents that are willing to bid more aggressively. Therefore, the Conservative Agent represents a risk-averse strategy that prioritises efficiency and profit margins over frequent wins [4].

C. Aggressive Agent

The Aggressive Agent follows a high-competition bidding strategy. It attempts to outbid other agents by making larger increases above the current highest bid. Compared with the Conservative Agent, it is more likely to win auctions because it places stronger bids and responds more competitively during each round.

This strategy is useful for testing whether frequent winning leads to better overall performance. However, a higher win rate does not always mean higher efficiency, especially in first-price auctions where the winner pays their own bid [3], [4]. Since the Aggressive Agent often bids closer to the perceived value of an item, it may reduce its potential profit margin. This means the agent can perform well in terms of winning auctions, but it may be less efficient if the cost of winning becomes too high.

D. Sniper Agent

The Sniper Agent uses a timing-based strategy. Instead of bidding throughout the whole auction, it waits until the later stage of the bidding process before placing a bid. This behaviour is based on the idea of entering the auction late to avoid early competition and reduce unnecessary bidding activity.

In the simulation, the Sniper Agent only participates after a certain proportion of the auction rounds has passed. When it does bid, it places a relatively large increase above the current highest bid. This gives the agent a chance to win late but also makes its performance highly dependent on timing and remaining bid opportunities. If the current price is already too high by the time the agent enters, it may fail to place a valid bid. As a result, the Sniper Agent represents a strategic but risky approach that depends heavily on auction conditions.

E. Adaptive Agent

The Adaptive Agent is the main learning-based strategy in the simulation, inspired by the idea that agents can improve decision-making through feedback from previous outcomes [8]. Unlike the other agents, it can use auction memory to adjust its bidding behaviour based on previous market outcomes. The memory stores information such as winning bids, losing bids, market prices, estimated market pressure, volatility, and competition level.

The Adaptive Agent begins with a base bid calculated from the perceived value of the item. This bid is then adjusted using several market-based factors. For example, if previous winning prices suggest a competitive market, the agent may increase its bid to improve its chance of winning. If previous wins suggest that bids are becoming too expensive, the agent can reduce its overpayment tendency. The agent also updates its aggressiveness depending on whether it has experienced more losing bids or winning bids.

This strategy allows the Adaptive Agent to respond to changing market conditions instead of following a fixed rule, which reflects the broader idea of adaptive behaviour in

intelligent agents [1], [8]. However, the quality of its performance depends on whether the memory information is useful. If the agent reacts too strongly to previous outcomes, it may overbid or become inefficient. Therefore, the Adaptive Agent is important for evaluating whether access to market information improves bidding performance, and whether learning behaviour always leads to better outcomes in a competitive auction environment.

V. IMPLEMENTATION

The implementation of this project consists of a fully simulated auction market in which multiple autonomous bidding agents compete for items over repeated auction rounds. The system was implemented in Python using an object-oriented structure, with separate classes used for the auction environment, auction items, agent behaviours, and market memory. This made the system easier to extend, as new agent strategies or experimental conditions could be added without rewriting the full simulation.

A. Auction Environment Design

The auction environment is responsible for managing the full simulation process. Each simulation consists of a fixed number of auction rounds. In every round, the environment generates an item, asks each agent to submit a bid, determines the winning bid, calculates the winner's profit, updates agent statistics, and stores the result.

Each auction follows a first-price auction structure where agents submit bids according to their own strategy, and the winning agent pays the submitted winning bid [3], [4]. However, during each round, the current highest bid is updated as agents act, meaning that later agents must exceed the current highest valid bid in order to remain competitive. To reduce ordering bias, the bidding order is randomly shuffled in each round.

The environment also validates whether bids are usable. A bid is only accepted if it is greater than the current highest bid, greater than zero, and within the agent's remaining balance. If no valid bids are submitted, the round ends without a winner. If multiple agents submit the same highest bid, the winner is selected randomly from the tied agents.

Once a winner is selected, the environment calculates the profit using:

$$\text{profit} = \text{item.true_value} - \text{winning_bid} \quad (1)$$

This means that profit is based on the actual value of the item rather than the agent's perceived value. This distinction is important in the market noise experiment, where agents may make decisions using noisy information but are still evaluated against the true value of the item.

B. Auction Items and Market Noise

Each auction item is represented using an AuctionItem class. The item stores two main values: the true_value and the perceived_value. Under normal conditions, these values are the same. However, in the market noise experiment, the perceived value is modified to simulate uncertainty in the market.

Market noise is applied at three levels: low, medium, and high. A low noise level applies a smaller random distortion to the item value, while a high noise level applies a larger

distortion. This allows the simulation to test whether agent performance changes when the information available to agents becomes less reliable.

The noise mechanism changes the value used by agents when deciding how much to bid. For example, an agent may perceive an item as being worth more or less than its actual value. However, the final profit calculation still uses the true value. This creates a more realistic distinction between decision-making information and actual market outcome.

This design was used to answer the question of how uncertainty affects agent behaviour. If an agent overestimates the item value, it may bid too aggressively and reduce its profit. If it underestimates the value, it may avoid profitable opportunities. Therefore, market noise tests not only performance, but also the robustness of each strategy.

C. Auction Memory

A memory component was implemented to store information from previous auction rounds. The Memory class records winning bids, losing bids, market prices, and round-level history. This allows agents with memory access to base their future decisions on previous auction outcomes, which is related to learning from feedback in intelligent agent systems [8].

The memory provides several market signals. These include the recent market average, market volatility, estimated market pressure, and estimated competition level. These values are calculated from previous winning bids and are used by the adaptive agent to adjust its bidding behaviour.

The memory system is especially important for Experiment 3, where the adaptive agent is tested with and without access to memory. This allows the project to evaluate whether additional market information improves performance. It also allows the adaptive agent to demonstrate a simple learning mechanism, since its behaviour changes based on previous wins and losses.

D. Implemented Agents

Five main agent strategies were implemented. Each agent extends from a shared BaseAgent class. The base class stores common information such as balance, total wins, total spent, total profit, failed bids, bid history, and profit history. It also provides shared methods for validating bids, recording bids, calculating win rate, and logging memory data.

a) Random Agent: Acts as a baseline strategy. It places a random bid between the minimum valid bid and a percentage of the perceived item value. This agent does not attempt to optimise its behaviour, making it useful as a comparison point for the more structured strategies.

b) Conservative Agent: Follows a low-risk bidding strategy. It only increases the current highest bid by a small amount and avoids bidding close to the full perceived value of the item. This agent is designed to protect profit margins, although it may lose more auctions because it does not bid aggressively.

c) Aggressive Agent: Bids more competitively by increasing the current highest bid by a larger amount. This gives it a higher chance of winning auctions, but it may reduce its profit by spending more. This strategy represents agents that prioritise winning over efficiency.

d) *Sniper Agent*: Only begins bidding in the later stage of a simulation round. Its purpose is to model a late-entry strategy where the agent avoids early competition and attempts to win near the end. This can sometimes reduce unnecessary bidding, but it may also miss opportunities if the current highest bid becomes too high before it acts.

e) *Adaptive Agent*: The most advanced strategy. It uses the perceived item value as a base, then adjusts its bid using memory-based market signals such as average price, volatility, pressure, competition, and recent overpayment behaviour. It also maintains an aggressiveness value, which changes over time depending on the balance between previous winning and losing bids. This allows the agent to increase or decrease its willingness to bid based on past market outcomes.

E. Simulation Runner and Result Logging

To support fair evaluation, a simulation runner was created to automate repeated experiment runs. Each experiment was run multiple times rather than relying on a single simulation, following the idea that agent performance in simulated markets should be evaluated across repeated trials rather than isolated outcomes [6], [7]. This reduces the effect of randomness and allows average performance and standard deviation to be calculated.

The system records several performance metrics for each agent. These include total profit, win rate, average bid, bid efficiency, failed bids, and total wins. Bid efficiency is calculated by comparing total profit against total spending. This is useful because an agent may achieve high profit but only by spending heavily, while another agent may achieve lower profit with better efficiency.

Results are exported into CSV files for analysis. Raw results are stored separately from summary results. The raw files contain individual run-level outcomes, while the summary files contain averages and standard deviations grouped by experiment condition. This structure makes it easier to analyse the results using pandas and create visualisations for the report.

The final analysis script imports the CSV files, computes comparison tables, and generates figures for the Results and Analysis section. These figures include bar charts for strategy comparison, line graphs for competition and market noise, error bars for variation across repeated runs, and a learning-over-time graph for the adaptive agent's aggressiveness.

VI. EXPERIMENTAL SETUP

The experiments were designed to evaluate how different autonomous bidding strategies behave under controlled auction conditions, similar to previous work where trading and bidding agents are tested in simulated competitive markets [6], [7]. Each experiment used repeated simulations so that the results were not based on a single random run. Since item values, bidding order, and some agent decisions involve randomness, each experiment was repeated multiple times, and the average results were recorded. Standard deviation was also calculated to show how stable or unstable the results were across repeated runs.

Each simulation consisted of 30 auction rounds. In each round, a new item was generated with a true value, agents submitted bids, the highest valid bidder won the item, and profit was calculated as the true item value minus the winning bid. Each agent started with the same initial balance to ensure

that the comparison was fair. The main performance measures used were total profit, win rate, average bid, bidding efficiency, and failed bids. Bidding efficiency was calculated by comparing the profit gained against the total amount spent, which helped show whether an agent was winning auctions profitably or simply spending more to win.

Fixed random seeds were used in the final experiment scripts to make the results reproducible across reruns. This ensured that the same experiment settings could regenerate the same summary results and figures, while repeated runs were still used to calculate averages and standard deviations.

A. Experiment 1: Strategy Comparison

The first experiment compared the performance of the main bidding strategies under the same auction conditions. The agents tested were the Random Agent, Conservative Agent, Aggressive Agent, Sniper Agent, and Adaptive Agent. The purpose of this experiment was to identify which strategy performed best in a standard market environment.

The independent variable was the agent strategy. The dependent variables were average profit, win rate, bidding efficiency, and failed bids. This experiment provided the baseline comparison for the rest of the study because it showed the general strengths and weaknesses of each strategy before changing other market conditions.

B. Experiment 2: Number of Agents and Competition

The second experiment tested how increasing the number of agents affected market performance. The number of agents was varied across three conditions: 5 agents, 10 agents, and 20 agents. This experiment was used to simulate different levels of competition within the auction market.

The aim was to observe whether agents could still make profitable decisions when the market became more crowded. As the number of agents increased, there were more bidders competing for the same items. This allowed the experiment to test whether higher competition reduced average profit, lowered bidding efficiency, or caused some strategies to become less effective.

C. Experiment 3: Information Availability

The third experiment focused on the Adaptive Agent and tested whether access to memory improved its performance. Two conditions were compared: the Adaptive Agent with memory access and the Adaptive Agent without memory access. In the memory condition, the agent could use previous auction information, such as winning bids, losing bids, market pressure, volatility, and competition estimates. In the no-memory condition, the agent could still bid, but it could not use previous market outcomes to adjust its decision-making.

The purpose of this experiment was to evaluate whether additional market information helped the Adaptive Agent make better bidding decisions, reflecting the role of feedback and state information in adaptive agent behaviour [1], [8]. The results were measured using profit, win rate, and efficiency. This experiment was important because it tested whether a more informed agent was better, or whether memory could sometimes lead to over-adjustment and inefficient bidding.

D. Experiment 4: Market Noise

The fourth experiment tested how uncertainty in item valuation affected agent performance. Market noise was applied by creating a perceived value for each item, which

could differ from its true value. Agents used the perceived value when deciding how much to bid, but final profit was still calculated using the true item value. This made the experiment more realistic because agents did not always have perfect information about the actual value of an item.

Three noise levels were tested: low, medium, and high. Low noise represented a small difference between perceived and true value, while high noise represented a more uncertain market. The purpose of this experiment was to examine whether agents remained profitable when item valuation became less reliable. It also helped show which strategies were more robust under uncertainty.

Overall, the four experiments were designed to test strategy performance, competitive pressure, information availability, and market uncertainty. Together, these experiments provide evidence for evaluating how autonomous agents behave in a simulated auction market and whether adaptive decision-making improves performance compared with simpler fixed strategies.

VII. RESULTS AND ANALYSIS

This section presents the results obtained from the simulated auction experiments. Each experiment was repeated over multiple runs to reduce the effect of randomness in item generation, bidding order and agent behaviour. The results are analysed using average profit, win rate, bidding efficiency and standard deviation. Profit measures how much value an agent gains after paying the winning bid, win rate measures how often an agent wins auctions, and efficiency compares profit against total spending.

A. Experiment 1: Strategy Comparison

The first experiment compared the performance of the five implemented bidding strategies under the same auction conditions. The purpose of this experiment was to evaluate how different agent designs behave in a shared auction environment when the number of agents and auction rounds remain fixed.

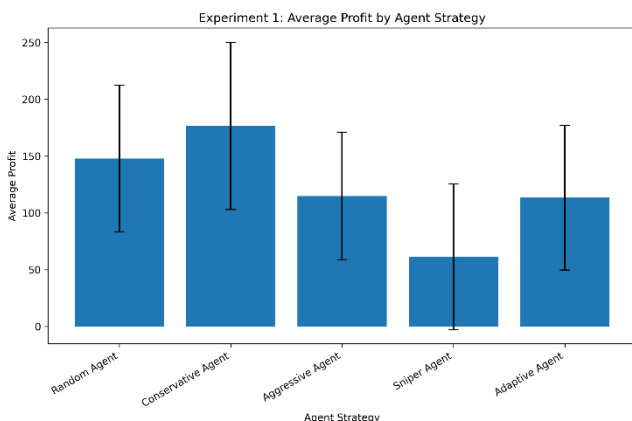


Fig. 1. Average profit achieved by each agent strategy across repeated simulation runs. Error bars represent standard deviation.

As shown in Figure 1, the Conservative Agent achieved the highest average profit overall, with an average profit of **176.60**. This suggests that cautious bidding was effective in this setting because the agent protected its profit margins while still winning enough auctions to perform well across repeated runs. The Random Agent achieved the second

highest average profit at **147.80**, followed by the Aggressive Agent at **114.90** and the Adaptive Agent at **113.50**.

The Conservative Agent's result shows that a low-risk strategy can be effective in a first-price auction because the winning agent pays its own bid. By avoiding unnecessarily high bids, the Conservative Agent was able to preserve more profit when it won. This also shows that the most competitive strategy is not always the most profitable one. Instead, profitability depends on how much value remains after paying the winning bid.

The Adaptive Agent did not achieve the highest average profit in this experiment, even though it used memory-based adjustment. Its average profit of **113.50** shows that adaptation did not automatically produce the best financial outcome. This suggests that the Adaptive Agent became more competitive, but also spent more aggressively to secure wins, which reduced its profit margins. The Aggressive Agent followed a similar pattern, where stronger bidding helped it compete but did not produce the highest profit.

The Sniper Agent performed the weakest, with an average profit of 61.40. This is likely because the Sniper Agent only participates in later bidding stages, meaning it had fewer opportunities to win auctions. Although late bidding can sometimes avoid unnecessary spending, the results show that waiting too long reduced its overall competitiveness in this simulation.

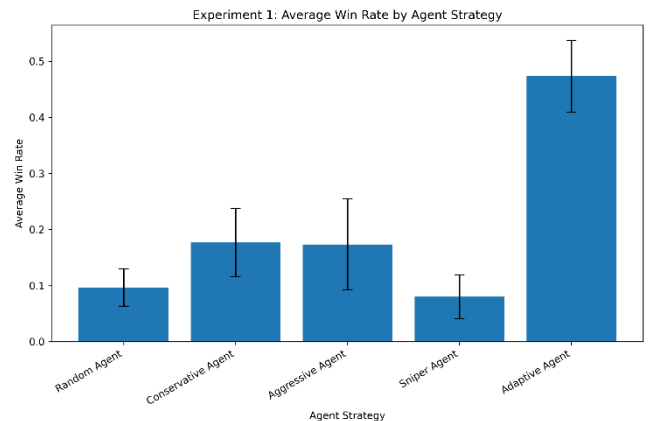


Fig. 2. Average win rate achieved by each agent strategy across repeated simulation runs. Error bars represent standard deviation.

Figure 2 shows that the Adaptive Agent achieved the highest average win rate at **0.4733**. This means it won auctions more frequently than all other strategies. However, this higher win rate did not lead to the highest average profit. This is an important finding because it shows that winning more auctions does not necessarily mean performing better in a first-price auction.

The Conservative Agent achieved a lower win rate of **0.1767** but still produced the highest average profit. The Aggressive Agent had a similar win rate of **0.1733**, while the Random Agent and Sniper Agent achieved lower win rates of **0.0967** and **0.0800** respectively. These results show that different strategies achieved success in different ways. The Adaptive Agent prioritised winning more auctions, while the Conservative Agent was more effective at preserving profit when it won.

This supports the trade-off described in auction theory, where bidders must balance the desire to win against the risk

of overpaying [3], [4]. In this experiment, the Adaptive Agent won most often, but its wins were not always economically efficient. Therefore, win rate alone is not enough to judge agent performance.

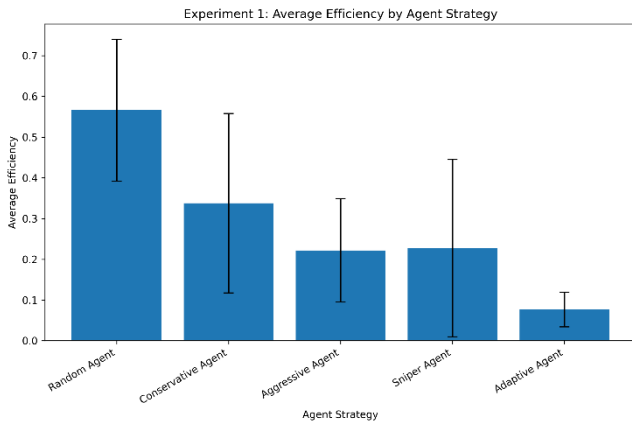


Fig. 3. Average bidding efficiency achieved by each agent strategy across repeated simulation runs. Error bars represent standard deviation.

Figure 3 shows that the Random Agent achieved the highest average bidding efficiency at **0.5662**. This does not mean that random bidding was the strongest overall strategy, because the Random Agent had a relatively low win rate. Instead, it suggests that when the Random Agent did win, it sometimes won auctions at relatively low prices and gained a high return compared with its spending.

The Conservative Agent also achieved a strong efficiency score of **0.3377**, supporting the idea that cautious bidding can preserve profit margins. The Sniper Agent achieved an efficiency of **0.2274** and the Aggressive Agent achieved **0.2219**. The Adaptive Agent had the lowest efficiency at **0.0766** despite having the highest win rate. This shows that the Adaptive Agent spent heavily to secure its wins, reducing the profit gained relative to the amount spent.

Overall, Experiment 1 shows that no single strategy dominated every metric. The Conservative Agent achieved the highest average profit, the Adaptive Agent achieved the highest win rate, and the Random Agent achieved the highest efficiency. This provides an important result for the project: successful bidding cannot be measured by win rate alone. In a first-price auction, an effective agent must balance competitiveness with controlled spending so that winning auctions still leads to meaningful profit.

B. Experiment 2: Number of Agents and Competition

The second experiment investigated how agent performance changed as the number of competing agents increased. The number of agents was varied across **5, 10 and 20 agents**, while the number of auction rounds remained fixed at **30**. This experiment was designed to test how market competition affects profitability, win rate and bidding efficiency.

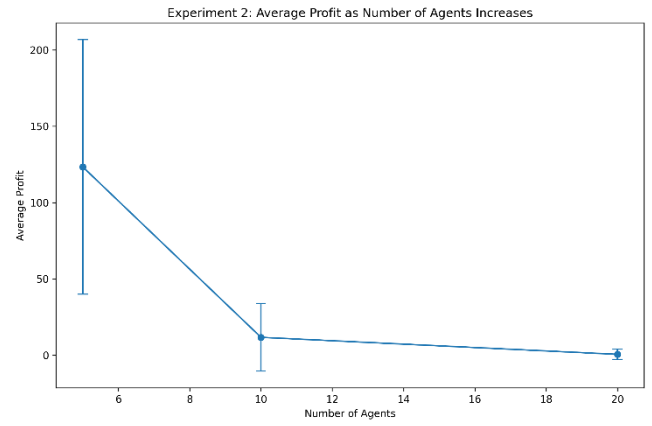


Fig. 4. Average profit as the number of agents increases from 5 to 20. Error bars represent standard deviation.

As shown in Figure 4, average profit decreased sharply as the number of agents increased. With **5 agents**, the average profit was **123.36**. When the number of agents increased to **10**, the average profit dropped to **11.71**. Under the highest competition setting of **20 agents**, the average profit fell further to only **0.56**.

This shows that increasing competition made it much harder for agents to generate profit. Since the number of auction rounds stayed fixed, more agents were competing for the same number of available items. This reduced each agent's opportunity to win and increased pressure to place higher bids. As a result, even when agents won auctions, the difference between the true item value and the winning bid became much smaller.

This result is consistent with the idea that multi-agent environments become more difficult when more agents compete for the same resources [2]. In a crowded auction market, the outcome of each agent is strongly affected by the presence and behaviour of other agents. Therefore, agent performance depends not only on the bidding strategy itself, but also on the level of competition in the environment.

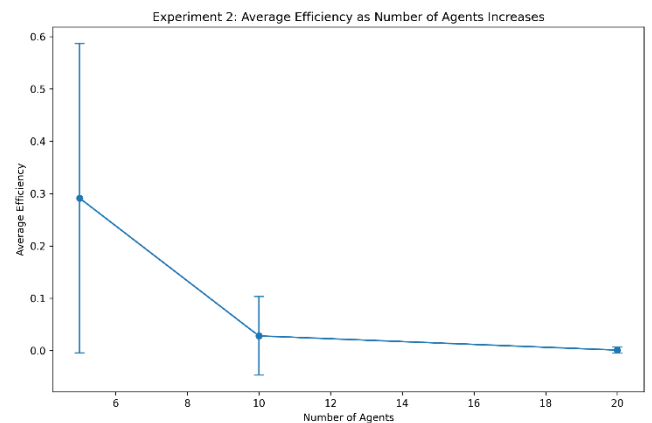


Fig. 5. Average bidding efficiency as the number of agents increases. Error bars represent standard deviation.

Figure 5 shows a similar decline in bidding efficiency. With **5 agents**, the average efficiency was **0.2912**. This decreased to **0.0282** with **10 agents** and collapsed to **0.0011** with **20 agents**. This suggests that agents were gaining much less profit relative to the amount they spent as the market became more crowded.

The win rate also decreased as expected. With **5 agents**, the average win rate was **0.2000**. With **10 agents**, it fell to **0.1000**, and with **20 agents**, it fell to **0.0500**. This is partly because more agents were sharing the same number of auction opportunities. However, the profit and efficiency results show that the effect was not only caused by fewer wins. The quality of each win also decreased because agents had to bid closer to item value to compete.

This shows that increasing competition affects both the frequency of winning and the profitability of winning. In a low-competition environment, agents can win items while still preserving a profit margin. In a high-competition environment, agents may need to spend so much to win that the final profit becomes very small.

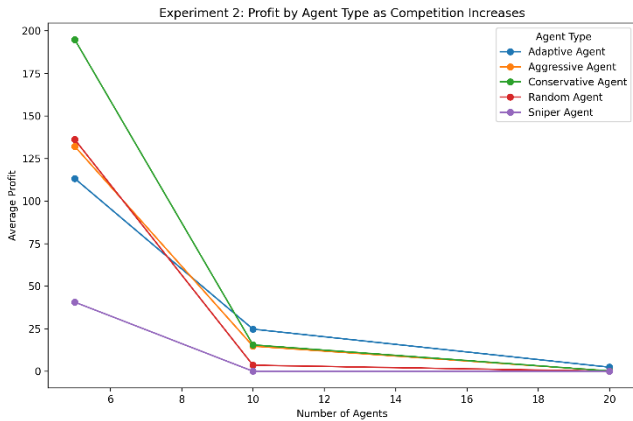


Fig. 6. Average profit by agent type as the number of agents increases.

Figure 6 shows that all agent types were negatively affected by increased competition. In the **5-agent** condition, agents were still able to generate meaningful profit because competition was limited and each agent had more opportunities to win. However, when the market increased to **10 agents**, profits dropped significantly across the environment. Under the **20-agent** condition, average profit was close to zero, showing that intense competition made it difficult for any strategy to perform well.

This result highlights an important limitation of simple bidding strategies. A strategy that performs reasonably well in a smaller market may not remain effective when the auction becomes crowded. Even adaptive or competitive agents can struggle when there are too many bidders competing for the same number of items.

Overall, Experiment 2 shows that competition has a strong negative effect on agent performance. As the number of agents increased, average profit, win rate and efficiency all decreased. This supports the conclusion that market structure is a major factor in autonomous bidding environments. Agent design is important, but even a strong strategy can lose effectiveness when the environment becomes too competitive.

C. Experiment 3: Information Availability

The third experiment investigated whether access to auction memory improved the Adaptive Agent's performance. Two conditions were tested: one where the Adaptive Agent had access to memory, and one where it did not. Memory allowed the agent to use previous winning bids, losing bids, market average, market volatility and competition pressure when deciding future bids.

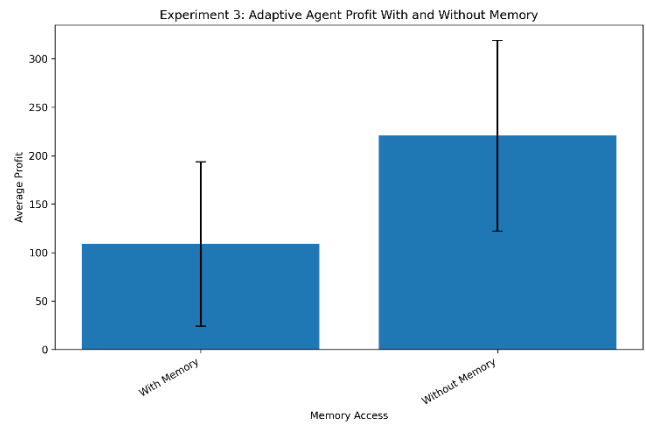


Fig. 7. Adaptive Agent average profit with and without memory access. Error bars represent standard deviation.

Figure 7 shows that memory access did not improve profitability. With memory, the Adaptive Agent achieved an average profit of **109.0**. Without memory, its average profit increased to **220.6**. This indicates that although the memory-based agent had access to additional market information, it did not always use this information in a profitable way.

The main reason for this result is that the memory-based agent became more competitive and placed higher bids to increase its chance of winning. This helped it win more auctions but also reduced the profit margin because the winning bid was often close to the item's value. In contrast, the no-memory version used a simpler bidding rule and won fewer auctions, but when it did win, it often spent less and therefore gained more profit.

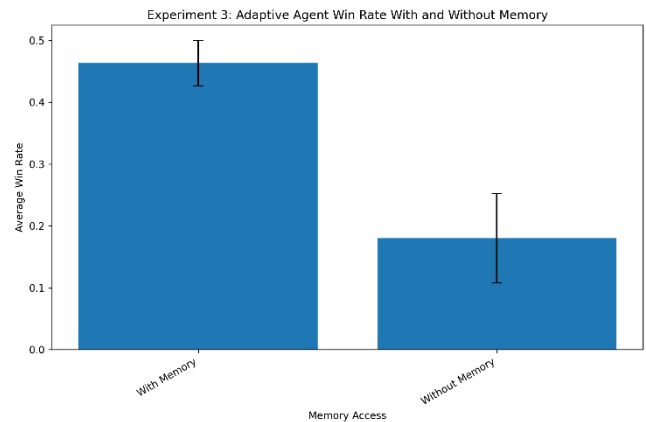


Fig. 8. Adaptive Agent average win rate with and without memory access. Error bars represent standard deviation.

Figure 8 shows that memory access did improve win rate. With memory, the Adaptive Agent achieved an average win rate of **0.4633**, compared with only **0.1800** without memory. This means the memory-based agent won more than twice as many auctions as the no-memory version.

However, this higher win rate did not translate into higher profit. This is an important finding because it shows that winning more auctions is not always beneficial in a first-price auction. Since the winner pays their own bid, an agent can win frequently but still perform poorly if it consistently overpays [3], [4]. Therefore, win rate alone is not enough to judge agent performance.

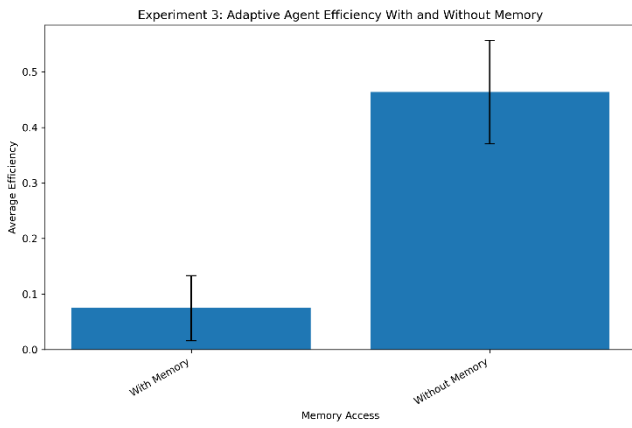


Fig. 9. Adaptive Agent average efficiency with and without memory access. Error bars represent standard deviation.

Figure 9 further supports this interpretation. With memory, the Adaptive Agent's average efficiency was only **0.0746**. Without memory, its efficiency was much higher at **0.4635**. This means the no-memory agent gained more profit for each unit spent, while the memory-based agent spent heavily for comparatively smaller returns.

These results show a trade-off between competitiveness and profitability. Memory access made the Adaptive Agent more successful at winning auctions, but it also made its bidding behaviour less efficient. The adaptive mechanism increased its aggressiveness when it experienced more losses than wins, which helped it compete, but also caused it to bid closer to the perceived item value. As a result, the agent prioritised winning rather than maximising profit.

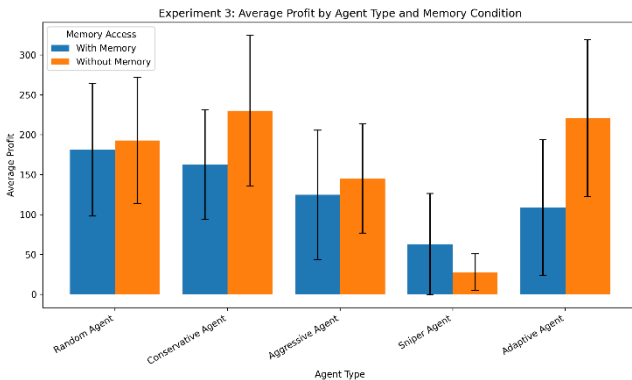


Fig. 10. Average profit by agent type under memory and no-memory conditions.

Figure 10 compares the wider market behaviour under both memory conditions. The Adaptive Agent had the highest average win rate when memory was enabled, but its profit was lower than in the no-memory condition. Other agents also changed performance depending on the memory condition because the Adaptive Agent's behaviour affected the overall auction environment. When the Adaptive Agent became more aggressive, it increased competitive pressure and changed the distribution of winning bids.

Overall, Experiment 3 produced a mixed result. Memory access improved the Adaptive Agent's ability to win auctions, but it reduced profitability and bidding efficiency. This suggests that additional information is only useful if the agent's decision rule uses it effectively. In this implementation, the memory mechanism helped the agent

respond to the market, but it also encouraged overbidding, making it less profitable overall.

D. Experiment 4: Market Noise

The fourth experiment evaluated how market noise affected agent performance. Market noise was introduced by changing the **perceived value** of each item while keeping the **true value** unchanged. Agents made bidding decisions using the perceived value, but profit was calculated using the actual true value. This allowed the experiment to test how agents behaved when their valuation of an item became less reliable, which is relevant to decision-making under uncertainty in intelligent agent environments [1], [2].

Three noise levels were tested: **low**, **medium**, and **high**.

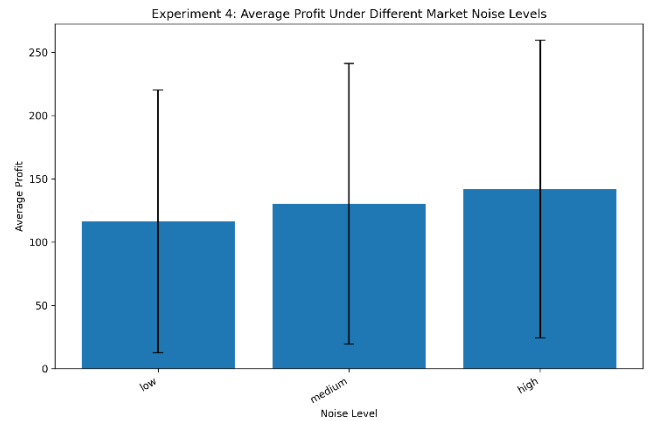


Fig. 11. Average profit under different market noise levels. Error bars represent standard deviation.

Figure 11 shows that average profit increased slightly as noise increased. Under low noise, the overall average profit was **116.46**. This increased to **130.38** under medium noise and **141.94** under high noise. This suggests that market noise did not simply reduce performance. Instead, it created more variation in bidding outcomes, allowing some agents to win auctions at lower bids and achieve higher profit margins.

However, the standard deviation also remained high across all three conditions, showing that performance became unstable. This means that although average profit increased, the results varied considerably between runs. Therefore, higher noise did not make the market consistently better for agents. It made the market less predictable.

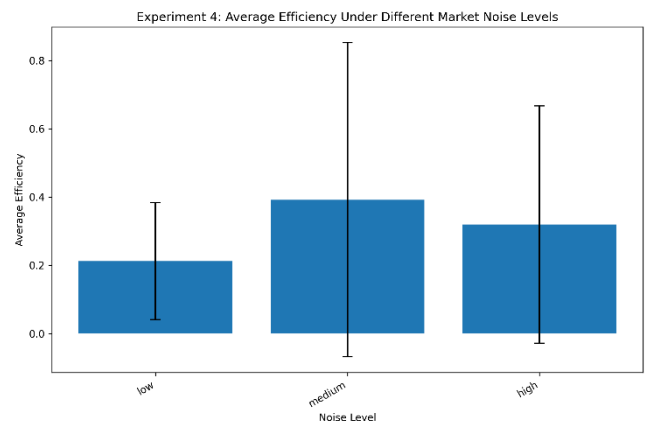


Fig. 12. Average efficiency under different market noise levels. Error bars represent standard deviation.

Figure 12 shows that bidding efficiency was highest under medium noise. The average efficiency increased from **0.2123** under low noise to **0.3919** under medium noise, before decreasing to **0.3199** under high noise. This suggests that a moderate level of uncertainty sometimes created profitable bidding opportunities, but excessive noise made decisions less reliable.

This result is important because it shows that uncertainty does not always have a linear effect. A small or moderate amount of noise may allow agents to exploit differences between perceived and true value. However, when noise becomes too high, agents are more likely to misjudge item value and either overbid or miss profitable opportunities.

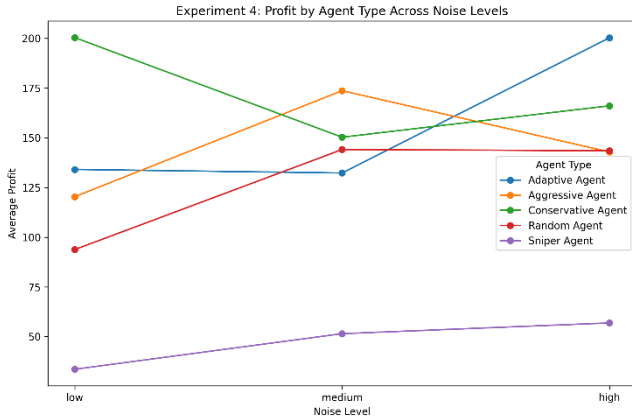


Fig. 13. Average profit by agent type across different market noise levels.

Figure 13 shows how each agent type responded to increasing noise. The Adaptive Agent performed strongly in terms of win rate across all noise conditions. Its average win rate was **0.4867** under low noise, **0.5033** under medium noise and **0.5100** under high noise. This indicates that the Adaptive Agent remained highly competitive even when item values became less certain.

In terms of profit, the Adaptive Agent achieved **134.1** under low noise, **132.3** under medium noise and **200.3** under high noise. This shows that the Adaptive Agent benefited most under high noise, likely because its aggressive bidding allowed it to capture more auctions when other agents were affected by uncertain valuations. However, its efficiency remained relatively low compared with some other agents, meaning it still spent heavily to achieve these wins.

The Conservative Agent performed well under low noise, achieving the highest average profit in that condition at **200.4**. This suggests that cautious bidding was effective when item values were relatively stable. However, its profit decreased under medium noise to **150.3** and only partially recovered under high noise to **166.1**. This indicates that conservative behaviour may become less reliable when the market becomes more uncertain.

The Aggressive Agent performed best under medium noise, achieving an average profit of **173.7** and relatively high efficiency. This suggests that aggressive bidding can benefit from uncertainty when it allows the agent to win valuable items before competitors adjust. However, under high noise, its average profit decreased to **142.9**, suggesting that excessive uncertainty made aggressive bidding riskier.

The Sniper Agent remained the weakest overall strategy in this experiment. Its average profit increased from **33.6** under low noise to **51.5** under medium noise and **56.9** under high noise, but it still performed below the other strategies. This suggests that bidding late was not enough to overcome uncertainty, especially when other agents had already pushed the current highest bid close to the perceived value.

Overall, Experiment 4 shows that market noise changed the auction environment in a meaningful way. Higher noise increased unpredictability and created more variation in agent outcomes. Some agents, especially the Adaptive Agent and Aggressive Agent, were able to benefit from this uncertainty in certain conditions. However, the high standard deviations show that noisy markets were less stable and less predictable. This supports the idea that uncertainty can create both opportunities and risks for autonomous bidding agents.

VIII. DISCUSSION

The experiments show that agent performance in the auction environment was strongly affected by both the bidding strategy used and the market conditions created in the simulation, which reflects the importance of environment design in multi-agent systems [1], [2]. Although all agents operated under the same auction rules, their outcomes differed because each strategy made different assumptions about risk, competition and available information. This demonstrates that in a multi-agent auction system, success is not only determined by winning items, but also by how efficiently an agent converts spending into profit.

A. Overall Strategy Performance

The results from Experiment 1 show that no single strategy dominated across every metric. The Conservative Agent achieved the highest average profit, suggesting that cautious bidding can be effective in a first-price auction because it protects profit margins. By avoiding unnecessarily high bids, the Conservative Agent was able to gain more value from the auctions it won.

The Adaptive Agent achieved the highest win rate, showing that its memory-based adjustment made it highly competitive. However, this did not lead to the highest profit or efficiency. This is one of the clearest findings from the simulation. Adaptive behaviour helped the agent win more auctions, but it also encouraged the agent to spend more aggressively. As a result, the Adaptive Agent often paid close to the item value, which reduced the profit gained from each win.

The Random Agent achieved the highest bidding efficiency. This does not mean that random bidding was the best strategy overall, because it had a lower win rate and less controlled behaviour. However, it shows that an agent can appear efficient if it wins fewer auctions but wins them cheaply. This reinforces the importance of using several evaluation metrics rather than judging performance from one measure alone.

The Aggressive Agent was competitive but did not achieve the highest profit. This is expected because its strategy places larger bids to increase the chance of winning. In first-price auctions, this can reduce profit margins because the winner pays their own bid [3], [4]. The Sniper Agent performed weakest overall because it entered late and had fewer valid

bidding opportunities. Although late bidding can sometimes be useful, it was not effective in this simulation environment.

Overall, the strategy comparison shows that successful bidding requires a balance between competitiveness and controlled spending. Winning frequently is useful only if the agent can still maintain profit after paying the winning bid. This finding supports the wider conclusion that autonomous bidding agents should be evaluated using profit, win rate, efficiency and stability together.

B. Effect of Competition

Experiment 2 showed a clear negative relationship between the number of agents and overall profitability. When the number of agents increased from 5 to 10 and then 20, average profit dropped sharply. This suggests that competition had a strong effect on the auction market. With more agents bidding for the same number of items, the chance of each individual agent winning decreased, while the pressure to place higher bids increased.

This result is important because it shows that an agent strategy that performs well in a small market may not remain effective in a crowded market, since multi-agent outcomes depend on the behaviour of other competing agents [2]. In the 5-agent condition, agents had more opportunities to win and could still make reasonable profits. However, in the 20-agent condition, the market became much more competitive, and average profit fell close to zero. This indicates that higher competition can cause profit margins to collapse.

The reduction in efficiency also suggests that agents were spending more relative to the value they gained. In a competitive auction, agents may need to bid closer to the item's value to win. This makes auctions harder to profit from because the difference between the true item value and the winning bid becomes smaller. Therefore, the results show that market structure is just as important as agent design. Even a strong strategy can struggle when the environment becomes too competitive.

C. Effect of Information Availability

Experiment 3 produced one of the most interesting findings. The Adaptive Agent with memory achieved a much higher win rate than the version without memory, but it had lower average profit and lower efficiency. This means that access to additional information did improve the agent's ability to win auctions, but it did not automatically improve its overall profitability.

This happened because the memory-based Adaptive Agent adjusted its aggressiveness based on previous wins and losses. When the agent experienced losses, it became more aggressive to improve its chances of winning future auctions. This made the agent more competitive, which explains the higher win rate. However, this also caused the agent to bid closer to the perceived value of the item. As a result, it often won auctions with a smaller profit margin.

This is a useful result because it shows that more information is not always beneficial unless it is used carefully, as adaptive behaviour depends on how feedback is interpreted and applied by the agent [8]. An adaptive agent needs to balance two goals: winning auctions and maintaining profit. In this simulation, the memory mechanism helped the agent compete, but it did not always help it make better economic decisions. Therefore, the result does not mean that memory is

ineffective. Instead, it suggests that the current learning rule is simple and may need further refinement.

The no-memory Adaptive Agent was less competitive but more efficient. Since it did not continuously increase aggressiveness based on previous losses, it avoided some of the overbidding behaviour seen in the memory condition. This allowed it to achieve higher average profit despite winning fewer auctions. This finding supports the idea that learning mechanisms must be evaluated carefully. A learning agent can appear successful based on win rate but still perform poorly when profit and efficiency are considered.

D. Effect of Market Noise

Experiment 4 examined how agents performed when item values became uncertain through market noise. The results showed that average profit changed across low, medium and high noise conditions. Higher noise did not simply reduce performance in a straight line. Instead, it created more variation in outcomes, with some agents benefiting from noisy conditions while others became less stable.

This occurred because agents made bidding decisions based on perceived value, while profit was calculated using the true value of the item. When perceived value was lower than true value, an agent could win the item cheaply and gain a higher profit. However, when perceived value was higher than true value, an agent risked overpaying and reducing profit. Therefore, market noise introduced both opportunity and risk.

The Adaptive Agent performed strongly in the high-noise condition in terms of average profit and win rate. This suggests that its memory-based behaviour helped it remain competitive in uncertain environments. However, its efficiency remained relatively low compared with some other outcomes, showing that it still tended to spend heavily to secure wins. This again highlights the trade-off between winning and bidding efficiently.

The presence of negative profit values in some individual runs also shows that noisy markets can punish poor bidding decisions. When agents rely on inaccurate perceived values, they may place bids that seem reasonable from their perspective but are actually too high compared with the true value. This makes market noise a useful experimental variable because it tests whether agents can remain robust under uncertainty.

E. Adaptive Behaviour and Learning

The Adaptive Agent demonstrates a simple form of learning through its aggressiveness adjustment, based on the principle that agents can adapt future actions using feedback from previous outcomes [8]. Rather than using a fixed bidding rule throughout the simulation, it changes its bidding behaviour based on previous outcomes stored in memory. This makes it different from the Random, Conservative, Aggressive and Sniper agents, which follow predefined strategies without adapting to the market.

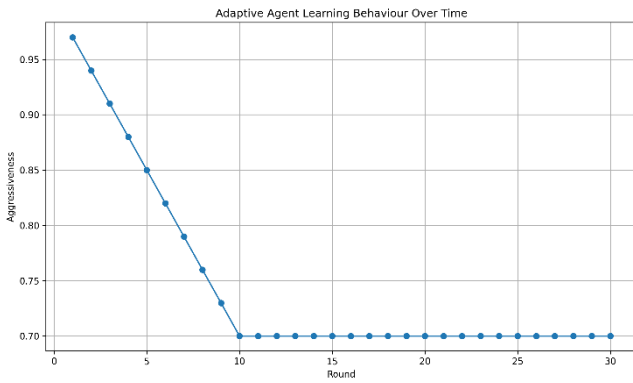


Fig. 14. Adaptive Agent aggressiveness over 30 auction rounds.

Figure 14 shows how the Adaptive Agent's aggressiveness changed over time. The agent began with an aggressiveness value of 0.97, then gradually reduced its aggressiveness during the early rounds. By round 10, it reached 0.70 and remained at this lower bound for the rest of the simulation. This shows that the agent was not static, because its internal bidding behaviour changed in response to previous auction outcomes. However, the stabilisation also shows that the learning rule was limited once the minimum aggressiveness value was reached.

This result is useful from an intelligent agent perspective because it shows how memory and feedback can influence decision-making. The Adaptive Agent adjusted its future bidding behaviour based on stored auction outcomes, rather than following a fixed rule throughout the simulation. However, the results also show that adaptation alone does not guarantee optimal performance. The Adaptive Agent often became more successful at winning auctions, but this sometimes came at the cost of efficiency.

This suggests that the learning rule was too focused on adjusting competitiveness after wins and losses, rather than explicitly optimising profit. A more advanced adaptive strategy could consider both previous wins and the profit gained from those wins. Overall, the Adaptive Agent was valuable because it revealed the tension between learning, competition and profitability. In auction-based multi-agent systems, intelligent behaviour should not only aim to win, but should also consider long-term profitability, risk and market uncertainty.

IX. CONCLUSION

A. Summary of Findings

This project investigated how different autonomous bidding agents perform in a simulated multi-agent auction market. The system was designed around repeated first-price auction rounds, where agents competed to win items while aiming to maximise profit. Several bidding strategies were implemented, including Random, Conservative, Aggressive, Sniper and Adaptive agents. The environment also included memory access, repeated simulations, market noise and different levels of competition to evaluate agent behaviour under changing conditions.

The results show that agent performance depends heavily on both the strategy used and the market environment. In the strategy comparison experiment, the Conservative Agent achieved the highest average profit at 176.60, while the Adaptive Agent achieved the highest win rate at 0.4733. This

shows that the most profitable agent was not necessarily the agent that won the most auctions. The Adaptive Agent was highly competitive, but its low efficiency showed that it often spent heavily to secure wins. This highlights an important trade-off in first-price auction environments: agents must balance winning frequently with maintaining profitable bids.

The information availability experiment further supported this finding. When memory was enabled, the Adaptive Agent achieved a higher win rate, but its average profit and efficiency decreased. With memory, it achieved an average profit of 109.00 and efficiency of 0.0746, while the no-memory version achieved an average profit of 220.60 and efficiency of 0.4635. This suggests that memory helped the agent become more competitive, but the decision rule did not always use that information in an economically efficient way. Therefore, additional information does not automatically improve performance unless the agent's adaptation mechanism is designed around the correct objective.

The experiments also showed that increased competition strongly reduced overall profit. As the number of agents increased from 5 to 10 and then 20, average profit dropped from 123.36 to 11.71 and then to 0.56. Bidding efficiency followed the same pattern, falling from 0.2912 to 0.0282 and then to 0.0011. This shows that crowded auction markets create stronger bidding pressure, reducing both the chance of winning and the profit margin available from each item.

The market noise experiment showed that uncertainty can create both opportunities and risks. Average profit increased from 116.46 under low noise to 141.94 under high noise, but the results were less stable because noisy perceived values could either help agents win cheaply or cause them to overpay. This shows that uncertain valuation does not always reduce performance directly, but it does make outcomes less predictable.

Overall, the project demonstrates that autonomous bidding agents can be evaluated effectively through simulation, which is consistent with previous work on autonomous bidding and trading agent competitions [6], [7]. The results support the idea that intelligent agent behaviour should not only be judged by success rate, but also by profitability, efficiency and robustness under uncertainty. The Adaptive Agent provided the most interesting behaviour because it changed its bidding behaviour based on previous outcomes, but the results also showed that its learning mechanism was still limited. A more advanced version could improve by learning from profit directly, rather than only reacting to wins and losses.

B. Reflection

This project helped demonstrate the importance of carefully designing both the agents and the environment in an intelligent agent system. At the beginning of the project, the main focus was on implementing different bidding strategies and comparing their outcomes. However, as the simulation developed, it became clear that the environment itself had a major influence on agent behaviour. Factors such as the number of competing agents, the availability of memory and the presence of market noise all changed the way agents performed.

One important lesson from the project was that a successful agent cannot be evaluated using only one metric. For example, the Adaptive Agent often achieved a high win rate, but this did not always mean it was the most profitable or

efficient agent. This showed the importance of using multiple evaluation measures, such as total profit, win rate, average bid, failed bids and efficiency. These metrics made it possible to analyse the agents more fairly and avoid drawing conclusions based only on how often an agent won.

The project also showed that simple adaptive behaviour can create meaningful results, even without using a complex machine learning model. The Adaptive Agent's memory-based adjustment was not perfect, but it created useful behaviour to analyse. This made the results more interesting because the agent did not simply perform better in every condition. Instead, it showed both strengths and weaknesses, which gave more depth to the discussion.

C. Future Work

There are several ways this project could be extended in the future. The most important improvement would be to develop a more advanced Adaptive Agent that learns from profit rather than only from wins and losses. In the current version, the agent increases or decreases its aggressiveness based on previous outcomes, but it does not directly consider whether those outcomes were economically efficient. A stronger version could use profit as part of its reward signal, allowing it to balance competitiveness with profitability.

Another possible extension would be to implement reinforcement learning, where an agent learns a bidding policy through repeated interaction and reward feedback [8]. This would allow the agent to learn a bidding policy over many auction episodes, rather than relying on manually defined adjustment rules. For example, the agent could learn when to bid conservatively, when to increase its bid and when to avoid an auction entirely. This would make the adaptive behaviour more realistic and would better connect the project to intelligent agent learning methods.

The auction environment could also be expanded to include more realistic market features. Future versions could include different auction formats, changing item supply, varying agent budgets, reserve prices, or agents with different risk preferences. The current simulation uses repeated independent auctions, but real markets often include changing conditions and strategic behaviour over time. Adding these features would make the environment more complex and closer to real-world automated bidding systems.

Finally, the evaluation could be improved by running a larger number of simulations and testing more market conditions. This would make the statistical analysis more reliable and reduce the effect of randomness. Additional visualisations could also be created to show changes in agent behaviour over time, such as bid trends, profit per round and balance changes. These extensions would provide a deeper understanding of how autonomous agents behave in competitive and uncertain auction environments.

REFERENCES

- [1] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Harlow, U.K.: Pearson, 2020.
- [2] Y. Shoham and K. Leyton-Brown, *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*. Cambridge, U.K.: Cambridge University Press, 2008.
- [3] W. Vickrey, "Counterspeculation, auctions, and competitive sealed tenders," *The Journal of Finance*, vol. 16, no. 1, pp. 8–37, Mar. 1961.
- [4] V. Krishna, *Auction Theory*, 2nd ed. Burlington, MA, USA: Academic Press, 2009.
- [5] R. H. Guttman, A. G. Moukas, and P. Maes, "Agent-mediated electronic commerce: A survey," *The Knowledge Engineering Review*, vol. 13, no. 2, pp. 147–159, 1998.
- [6] M. P. Wellman, A. Greenwald, P. Stone, and P. R. Wurman, "The 2001 Trading Agent Competition," *Electronic Markets*, vol. 13, no. 1, pp. 4–12, 2003.
- [7] M. P. Wellman, A. Greenwald, and P. Stone, *Autonomous Bidding Agents: Strategies and Lessons from the Trading Agent Competition*. Cambridge, MA, USA: MIT Press, 2007.
- [8] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.